

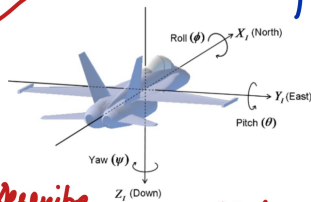
Drone Control Loop

2



* **Throttle**: The current, real-time number the Remote Controller (RC) sends the Flight Controller (FC), comes from joystick.

Let's say range from 0 (motor off) to 1000 (full power)



Describe 3D orientation of object with respect to fixed reference frame



Example:

Drone Flying → Wind gust → Unstable! → PID
To stabilize: Push left up, drop right down

Left motors spin faster, Right slower

So $\text{motor_left} = \text{throttle} + \text{roll correction}$
 $\text{motor_right} = \text{throttle} - \text{roll correction}$

Assuming all motors spin in the same direction



Motor PWM

① Using Electronic Speed Controllers (ESCs) → expect a pulse every 20ms.

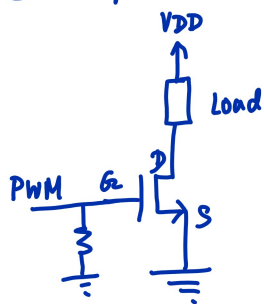
* Pulse of 1ms = 0% Throttle

* Pulse of 2ms = 100% Throttle

External ESC → may add weight!

So, take throttle (0-1000) → map to pulse width (1ms - 2ms) → Set PWM pulse

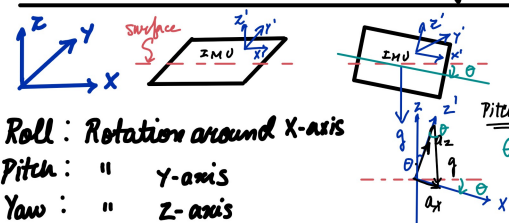
② Simple low side switching (+ flyback diode connected across motor load (inductive) to prevent damage from voltage spikes)



So, can map motor left & motor right directly to PWM percentage

* Note: Choose high PWM frequency to reduce vibrations ($\approx 20\text{ kHz}$)

Calculate Roll and Pitch angles



Roll: Rotation around X-axis

Pitch: " Y-axis

Yaw: " Z-axis

Similarly, Roll:

$$\phi = \tan^{-1} \left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}} \right)$$

$$\theta = \tan^{-1} \left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}} \right)$$